

Introduction

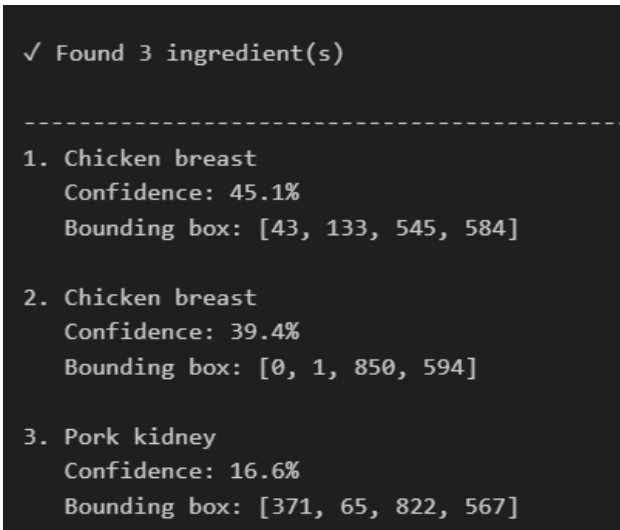
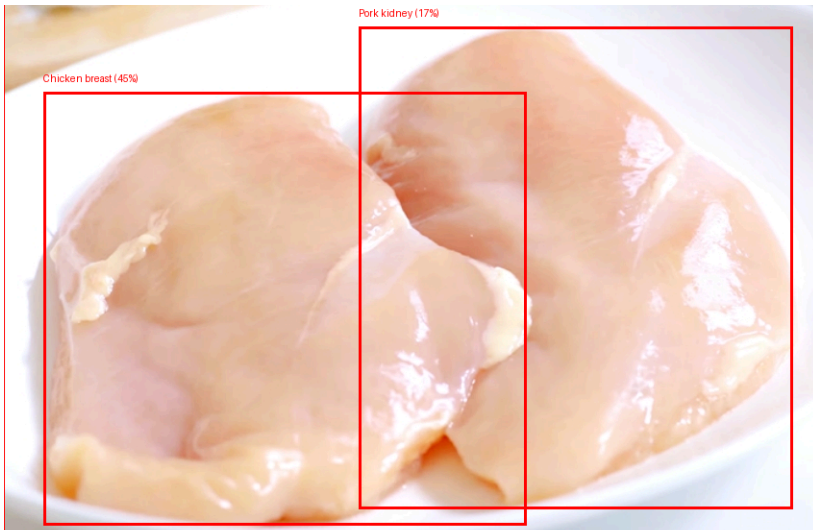
- Recipe generation from ingredient photos is an emerging application at the intersection of computer vision, nutrition analysis, and large language models.
- Many pipelines lack calorie estimation or structured validation, leading to inconsistent recipe quality and incorrect nutrition outputs.
- The system integrates CLIP + DETR for multi-ingredient detection, USDA nutrition lookup for calorie estimation, and multiple LLMs (GPT-2, Llama 1B, Llama 8B) for diverse, high-quality recipe generation.

Problem to Solve?

- How can we reliably detect multiple ingredients from a single food image using existing state-of-the-art vision models?
- Can we estimate calories and nutritional information from image-derived bounding boxes in a consistent and scalable way?
- How can we generate high-quality recipes that include realistic preparation time, cooking time, servings, and formatted steps?
- How do we ensure that generated recipes remain consistent, diverse, and robust across different LLM sizes and architectures?

Ingredient Detection

- Uses CLIP (ViT-Base) for zero-shot ingredient classification
- Uses DETR (ResNet-50) for object detection and bounding boxes
- Supports a vocabulary of 525+ ingredients
- Applies confidence-based filtering for more reliable detections
- Consolidates overlapping detections into a unified ingredient list
- Outputs ingredient names, bounding boxes, and confidence scores



Nutrition Estimation

- Utilizes the USDA FoodData Central database containing 525+ ingredients
- Maps detected ingredients to their closest nutritional entries using fuzzy matching
- Estimates ingredient weight based on detected bounding box area
- Computes calories, macronutrients, and basic nutrition per ingredient
- Aggregates all ingredients into a unified per-serving nutrition profile
- Ensures consistent, scalable nutrition estimation without manual labeling

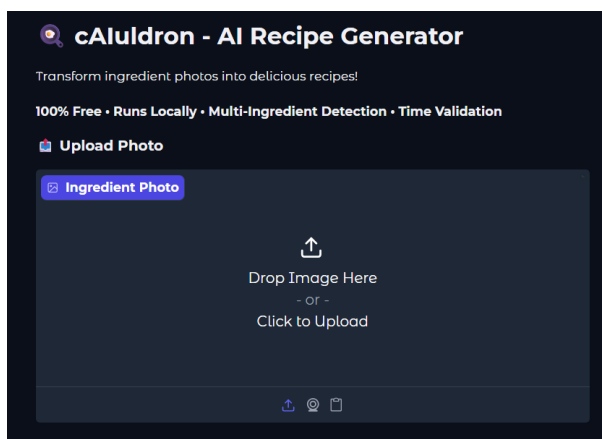
Recipe Generation

Model	Speed	Quality	VRAM Usage	Best Use Case
GPT-2 (Fine-tuned)	⚡⚡⚡	★	1-2 GB	Quick generation / low compute
Llama 3.2 1B (LoRA 4-bit)	⚡⚡	★★	4-5 GB	Recommended balance of speed & quality
Llama 3.1 8B (GGUF)	⚡	★★★	4-6 GB	Best recipe quality & detail

Model	Learning Rate	Batch Size	Epoch	Max Seq Length	Optimizer	Weight Decay
GPT-2 Fine-Tuning	0.00005	16	3	512	AdamW	0.01
Llama 1B (LoRA, 4-bit)	0.0002	8	3	1024	AdamW	0

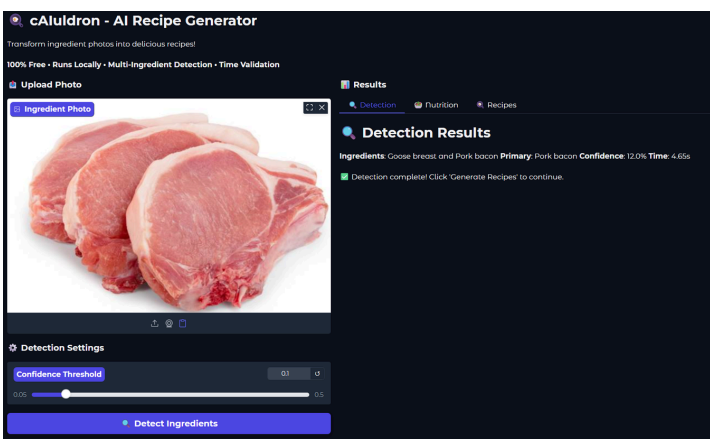
Pipeline&Output

Step1: User uploads a food / ingredient photo as the input.



Input: Ingredient photo
Model: CLIP + DETR
Output: Ingredient list

Step2: Use CLIP + DETR to detect ingredients, then output an ingredient list.

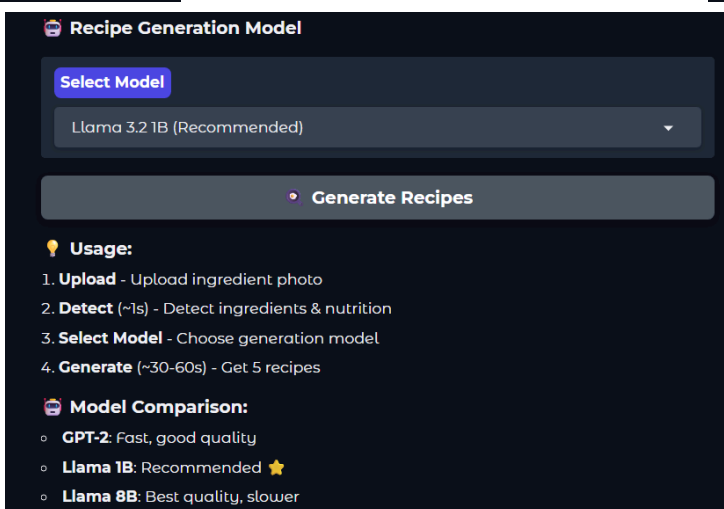


Step 3- Nutrition Estimation
Map detected ingredients to USDA FoodData Central, estimate weight from bounding boxes, and compute calories & macros.



Input: Ingredient list
Model: USDA Nutrition estimation
Output: Nutrition info (calories & macros)

After the detect the ingredient
Choose one generate model



Input: Ingredients + nutrition info
Model: generate model
Output: Full recipe (title, steps, ingredients, time fields)



Step 4- Recipe Generation
Generate a complete recipe using GPT-2 / Llama 1B / Llama 8B, including ingredients, steps, and time fields.

Step 5: Recipe Done
Let's Get Cooking and Bring Your Meal to Life!

